Trap Spaces

A **subspace** of $\mathbb{B}^n$ is a partial function $d\colon \{1,\ldots,n\} \hookrightarrow \mathbb{B}$.
The variables on which d is defined, $D = \mathit{range}(d) \subseteq \{1,\ldots,n\}$, are called **fixed**.
The variables which are not fixed, $F = \{1,\ldots,n\} \setminus D$, are called **free**.
We denote the subspaces by vectors $\mathbf{d} \in \{0,1,*\}^n$ such that for each $i \in \{1,\ldots,n\}$:

$$\mathbf{d}_i \triangleq \begin{cases} 0 & \text{if } i \in D \text{ and d(i)=0} \\ 1 & \text{if } i \in D \text{ and d(i)=1} \\ * & \text{if } i \in F \end{cases}$$

By abuse of notation, a subspace is also a set of configurations it contains, $\mathbf{d} = \{\mathbf{x} \in \mathbb{B}^n \mid \forall i \in D, \mathbf{d}_i = \mathbf{x}_i\}$.

A **trap space** is a subspace which is also a trap set.

$\forall x \in d \;,\; f(x) \in d$